

mooreVIEW Cloud

Functional UI Validation Checklist

Suite version: 2.3.8

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Deployment mode: Cloud SaaS (`MOOREVIEW_DEPLOYMENT=c1oud` , port 3100)

Complete end-to-end UI test plan for authentication, Studio, fleet management, CMMS, and standalone tools.

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mooreVIEW Cloud — Functional UI Validation Checklist

Document version: 1.0
Product: mooreVIEW Cloud Studio (SaaS deployment)
Scope: End-to-end UI functional validation for cloud mode (`MOOREVIEW_DEPLOYMENT=ccloud` , port 3100)

How to use this checklist

- 1. Deploy or start cloud SaaS locally: `npm run start:saas` → <http://127.0.0.1:3100>
- 2. Seed demo data if needed: `npm run seed`
- 3. Record session details in **Test session** below
- 4. Work top-to-bottom; mark each item **Pass**, **Fail**, or **N/A**
- 5. Log defects with the item ID (e.g. `AUTH-03`) in your tracker
- 6. Re-test failed items after fixes before sign-off

Legend: ☐ = not tested · ☒ = pass · ☒ = fail · ☐ = N/A

Test session

Field	Value
Tester	
Date	
Environment	☐ Local dev · ☐ Staging · ☐ Production
URL	
Build / version	
Browser(s)	☐ Chrome · ☐ Edge · ☐ Firefox · ☐ Safari
Viewport(s)	☐ Desktop (≥1280px) · ☐ Tablet (~768px) · ☐ Mobile (≤480px)

Test accounts

Role	Org ID	Email	Password	Used
Platform admin	demo	admin@demo.local	ChangeMeAdmin!	☐
Tenant admin / operator	demo	operator@demo.local	demo	☐
Custom tenant user				☐

1. Authentication & session

ID	Test case	Steps	Expected result	Result	Notes
AUTH-01	Login page loads	Navigate to /login while logged out	Split hero + sign-in card; fields: Organization ID, Email, Password; demo credentials hint visible	☐	
AUTH-02	Valid login (operator)	Enter demo / operator@demo.local / demo ; submit	Redirect to / (Studio); session cookie set; no error banner	☐	
AUTH-03	Valid login (platform admin)	Log out; log in as admin@demo.local	Redirect to Studio; Admin nav link visible	☐	
AUTH-04	Invalid credentials	Wrong password or unknown email	Error message shown; remain on /login ; no redirect	☐	
AUTH-05	Invalid org ID	Unknown organization slug	Error message; remain on login page	☐	
AUTH-06	Empty field validation	Submit with one or more fields blank	Inline or banner validation; form not submitted	☐	
AUTH-07	Protected route redirect	While logged out, open /sites	Redirect to /login?next=... (or equivalent)	☐	
AUTH-08	Post-login redirect (next)	Log in from /login?next=/fleet	Land on /fleet after successful auth	☐	
AUTH-09	Sign out — Studio topbar	Click Sign out in cloud nav or Help menu	POST logout; redirect to /login ; protected routes blocked	☐	
AUTH-10	Sign out — Cloud Studio pages	From /sites , click Sign out	Session cleared; / requires login again	☐	
AUTH-11	Session persistence	Log in; close tab; reopen app within 7 days	Still authenticated (or re-prompt per TTL policy)	☐	
AUTH-12	Login responsive layout	Resize to ≤860px width	Single-column layout; hero collapses; form usable	☐	

2. Studio shell & navigation

ID	Test case	Steps	Expected result	Result	Notes
SHELL-01	Studio page title	Open / as cloud user	Title shows "MooreVIEW Cloud Studio" (or cloud branding)	☐	
SHELL-02	Cloud secondary nav visible	On Studio /	Nav bar: Studio, Sites, Devices, Assets, People, CMMS; user label; Sign out	☐	
SHELL-03	Admin nav — platform admin	Log in as platform admin	Admin link visible in cloud nav	☐	
SHELL-04	Admin nav — operator hidden	Log in as operator (non-admin)	Admin link NOT visible	☐	
SHELL-05	Cloud nav — Studio link	Click Studio from /sites	Navigate to /	☐	
SHELL-06	Cloud nav — Sites link	Click Sites from Studio	Navigate to /sites	☐	
SHELL-07	Cloud nav — Devices link	Click Devices	Navigate to /sites/devices	☐	
SHELL-08	Cloud nav — Assets link	Click Assets	Navigate to /fleet	☐	
SHELL-09	Cloud nav — People link	Click People	Navigate to /people	☐	
SHELL-10	Cloud nav — CMMS link	Click CMMS	Navigate to /cmms	☐	
SHELL-11	Topbar — Project menu	Open Project ▾	Status, System setup, New/Open/Save/Import/Export/Delete, Deploy, Share, Save workspace	☐	
SHELL-12	Topbar — Historian menu	Open Historian ▾	Historian trend, Logger config, PdM entries open popups	☐	
SHELL-13	Topbar — Reporting menu	Open Reporting ▾	Reports and ROI calculator entries work	☐	
SHELL-14	Topbar — PdM shortcut	Click PdM	PdM asset setup popup opens	☐	
SHELL-15	Topbar — Alarms shortcut	Click Alarms	Alarms popup opens; badge shows count if active	☐	
SHELL-16	Topbar — CMMS shortcut	Click CMMS in topbar	Navigate to /cmms	☐	
SHELL-17	Topbar — MV Draw	Click MV Draw	Navigate to /mv-draw in same or new tab	☐	
SHELL-18	Topbar — Tools menu	Open Tools ▾	Program, Tags, Drivers, Connectivity (/cellular/sims)	☐	
SHELL-19	Topbar — Camera menu	Open Camera ▾	Camera admin popup + live camera shortcuts	☐	
SHELL-20	Topbar — Help menu	Open Help ▾	Training (F2), Help (F1), Sign out	☐	

SHELL-21	About popup	Click About	Version / product info popup displays	☐	
SHELL-22	F1 Help shortcut	Press F1 on Studio	Help popup opens with topic sidebar	☐	
SHELL-23	F2 Training shortcut	Press F2	Training curriculum popup opens	☐	
SHELL-24	Mobile topbar menu	Viewport ≤720px or landscape mobile	Hamburger / mobile menu exposes topbar items	☐	
SHELL-25	PWA manifest	Inspect page head	<code>manifest.webmanifest</code> , theme-color, apple-touch-icon present	☐	

3. HMI live display & composer

ID	Test case	Steps	Expected result	Result	Notes
HMI-01	Live HMI renders	Open project with HMI on Studio	Composed screen visible; tags update on poll	☐	
HMI-02	Screen navigation bar	Click alternate HMI screen tab	Screen switches; bindings refresh	☐	
HMI-03	Setup composer	Click Setup... on HMI panel	HMI composer popup opens (hmi-setup)	☐	
HMI-04	Composer — tile grid	Add/move/resize tiles in composer	Canvas updates; preview reflects changes	☐	
HMI-05	Composer — tag bindings	Bind a tag to a graphic element	Binding saved; live value appears on Apply	☐	
HMI-06	Composer — Apply	Click Apply in composer	Live HMI reflects changes without full reload	☐	
HMI-07	Reload screen	Click Reload screen	HMI refreshes from server state	☐	
HMI-08	Zoom controls	Use zoom in / out / fit	HMI viewport scales correctly	☐	
HMI-09	Test mode (technician)	Enable Test mode checkbox	10s refresh behavior active (if entitled)	☐	
HMI-10	Open 3D	Click Open 3D (if project has 3D)	3D view opens or navigates correctly	☐	
HMI-11	Alarm status button	Click alarm status on HMI chrome	Opens alarms or shows alarm summary	☐	
HMI-12	Room popup modal	Trigger room hotspot on HMI	Room detail modal opens with layout + condenser	☐	
HMI-13	Camera popup modal	Trigger camera hotspot on HMI	Camera iframe modal opens with live stream	☐	
HMI-14	HMI on narrow viewport	Resize to ≤720px	Live stage min-height preserved; controls stack; touch-friendly	☐	

4. Engineering tools (popups)

4a. Program (ST editor)

ID	Test case	Steps	Expected result	Result	Notes
PROG-01	Open Program popup	Tools → Program	ST editor popup opens with current program	☐	
PROG-02	Validate	Click Validate	Validation result shown (pass or error list)	☐	
PROG-03	Start runtime	Click Start	Runtime starts; status indicator updates	☐	
PROG-04	Pause runtime	Click Pause	Runtime pauses	☐	
PROG-05	Stop runtime	Click Stop	Runtime stops cleanly	☐	
PROG-06	Remote Opta checkbox	Toggle remote execution option	Remote mode engages when Opta connected	☐	
PROG-07	Connect / Disconnect	Use Connect/Disconnect for remote Opta	Connection state reflected in UI	☐	

4b. Tags

ID	Test case	Steps	Expected result	Result	Notes
TAG-01	Open Tags popup	Tools → Tags	Tag database table loads	☐	
TAG-02	Edit tag value	Change tag property; Save	Change persists after reload	☐	
TAG-03	Scale / alarm config	Edit scaling or alarm limits	Saved and reflected in live I/O	☐	
TAG-04	Force I/O	Force an output tag	Forced value shown; release works	☐	
TAG-05	Live I/O link	Open Live I/O while runtime running	Live program I/O popup shows current values	☐	
TAG-06	Ensure motor/TPO tags	Use ensure-tags action if present	Required tags created without error	☐	

4c. Drivers

ID	Test case	Steps	Expected result	Result	Notes
DRV-01	Open Drivers popup	Tools → Drivers	Driver list and templates visible	☐	

DRV-02	Add driver	Add new driver instance	Appears in list; config panel opens	☐	
DRV-03	Save driver config	Edit and Save	Config persists	☐	
DRV-04	Hardware wizard	Launch Hardware wizard	Guided template flow completes	☐	
DRV-05	Apply device template	Select template → Apply	Tags/I/O created per template	☐	
DRV-06	Bulk Opta Parc	Run bulk Opta Parc action	Devices discovered or listed	☐	
DRV-07	Modbus RTU↔TCP	Open Modbus move tool	Register migration UI functional	☐	
DRV-08	NextCentury API	Configure NextCentury credentials	Save succeeds; deploy estimate available	☐	

4d. Live I/O

ID	Test case	Steps	Expected result	Result	Notes
IO-01	Live I/O popup	Open from Tags or Tools	I/O table updates while runtime runs	☐	

5. Operations (alarms, historian, reports, cameras)

5a. Alarms

ID	Test case	Steps	Expected result	Result	Notes
ALM-01	Alarms popup list	Open Alarms; trigger test alarm	Alarm appears in active list	☐	
ALM-02	Ack single alarm	Click Ack on one alarm	Alarm acknowledged; state updates	☐	
ALM-03	Ack all	Click Ack all	All active alarms acknowledged	☐	
ALM-04	Alarm badge count	Active alarms exist	Topbar badge shows correct count	☐	
ALM-05	Alarm notify config	Open alarm-notify popup	User/recipient config saves	☐	

5b. Historian & PdM

ID	Test case	Steps	Expected result	Result	Notes
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HIST-01	Historian trend	Historian → trend popup	Chart renders with pen data	☐	
HIST-02	Clear graph	Click clear in historian	Chart resets	☐	
HIST-03	Logger config	Historian → Logger config	MongoDB logger settings apply	☐	
HIST-04	PdM setup tab	Historian logger → PdM	Asset setup, simulation controls work	☐	
HIST-05	PdM asset popup	Topbar PdM	Asset list and detail panels load	☐	
HIST-06	PdM PDF download	Download PdM asset PDF	PDF file downloads with asset data	☐	

5c. Reports

ID	Test case	Steps	Expected result	Result	Notes
RPT-01	Reports popup	Reporting → Reports	Report templates and pen config visible	☐	
RPT-02	CSV export	Generate CSV report	File downloads with expected columns	☐	
RPT-03	PDF export	Generate PDF report	PDF downloads successfully	☐	
RPT-04	ROI calculator	Reporting → ROI	ROI form calculates and displays results	☐	

5d. Cameras (Studio popup)

ID	Test case	Steps	Expected result	Result	Notes
CAM-01	Cameras popup tabs	Camera → admin	Tabs: Overview, Inventory, Detail, Archive, Events, AI history, Discover, System, Settings	☐	
CAM-02	Live camera shortcut	Camera menu → live entry	Stream iframe loads or shows offline state	☐	
CAM-03	Camera discovery	Discover tab → scan	Cameras found or empty state shown	☐	
CAM-04	Camera settings	Settings tab — go2rtc / AI	Settings save without error	☐	

6. Project & system setup

ID	Test case	Steps	Expected result	Result	Notes
PROJ-01	Status popup	Project → Status	Project status summary displays	☐	

PROJ-02	System setup — General	Project → System setup → General	Scan interval, HMI start screen, startup mode fields editable	☐	
PROJ-03	MQTT Parc hub config	General tab — MQTT section	Enable, broker URL, credentials, site key save	☐	
PROJ-04	Cloud remote uplink	General tab — cloud uplink	Tenant ID, gateway ID, broker credentials save	☐	
PROJ-05	Apply all settings	Click Apply in System setup	Settings persist; confirmation or no error	☐	
PROJ-06	Area configuration tab	System setup → Area config	Assisted living / pool site options save	☐	
PROJ-07	New project	Project → New	New blank or template project created	☐	
PROJ-08	Open project	Project → Open	Project list; selection loads project	☐	
PROJ-09	Save project	Project → Save	Project saved to server/storage	☐	
PROJ-10	Import / Export	Import .est / Export .est.zip	Round-trip succeeds	☐	
PROJ-11	Delete project	Delete with confirmation	Project removed from list	☐	
PROJ-12	Deploy project	Project → Deploy	Deploy flow completes or shows actionable errors	☐	
PROJ-13	Share project	Project → Share	Share dialog with local/cloud tabs	☐	
PROJ-14	Save workspace	Project → Save workspace	workspace.est.json saved	☐	
PROJ-15	Print project config	Projects tab → print	Print preview or PDF of configuration	☐	

7. Cloud fleet — Sites

ID	Test case	Steps	Expected result	Result	Notes
SITE-01	Sites page loads	Navigate to <code>/sites</code>	Sites table, Create form, Refresh button visible	☐	
SITE-02	Create site	Enter Site ID + Name; Create	Site appears in table; pairing code shown	☐	
SITE-03	Pairing code display	After create or select site	Pairing code visible in banner/row	☐	
SITE-04	Agent online badge	Site with connected agent	Online badge green / active	☐	
SITE-05	Agent offline badge	Site without agent	Offline or inactive badge	☐	
SITE-06	Camera count column	Site with synced cameras	Count matches <code>/sites/:id/cameras</code>	☐	

SITE-07	Refresh sites	Click Refresh	Table reloads from API	☐	
SITE-08	Delete site	Delete with confirm dialog	Site removed; confirm required	☐	
SITE-09	Cameras link	Click Cameras for a site	Navigate to <code>/sites/:siteId/cameras</code>	☐	
SITE-10	Empty state	New tenant with no sites	Helpful empty state; create form prominent	☐	

8. Cloud fleet — Site cameras

ID	Test case	Steps	Expected result	Result	Notes
SCAM-01	Cameras page loads	Open <code>/sites/:id/cameras</code>	Agent status badge; camera table (ID, name, model, probe)	☐	
SCAM-02	Open live player	Click Open live on a camera	Iframe player opens with stream or offline message	☐	
SCAM-03	Agent status accuracy	Compare agent badge to edge state	Badge matches actual agent connectivity	☐	
SCAM-04	Empty camera list	Site with no cameras synced	Empty table with clear messaging	☐	

9. Cloud fleet — Devices

ID	Test case	Steps	Expected result	Result	Notes
DEV-01	Devices page loads	Navigate to <code>/sites/devices</code>	All-org devices table visible	☐	
DEV-02	Add device	Fill Name, Site, Serial; submit	Device appears in table	☐	
DEV-03	Online status column	Device with telemetry	Online indicator correct	☐	
DEV-04	Commissioning column	New device	Commissioning state shown	☐	
DEV-05	Refresh devices	Click Refresh	Table reloads	☐	
DEV-06	Tenant isolation	Log in as different tenant	Only own-org devices visible	☐	

10. Cloud fleet — Assets map

ID	Test case	Steps	Expected result	Result	Notes
FLEET-01	Fleet page loads	Navigate to <code>/fleet</code>	Visual map + asset table visible	☐	
FLEET-02	Add asset	Enter Name, Site, Status, X/Y%; submit	Dot appears on map; row in table	☐	
FLEET-03	Status color coding	Set statuses: normal, alarm, offline	Map dots use distinct colors	☐	
FLEET-04	Map dot position	Set X/Y percentages	Dot positioned correctly on map	☐	
FLEET-05	Delete asset	Delete with confirm	Asset removed from map and table	☐	
FLEET-06	Refresh / reload	Reload page	Assets persist from API	☐	

11. People (tenant user management)

ID	Test case	Steps	Expected result	Result	Notes
PPL-01	People page loads	Navigate to <code>/people</code>	Users table + Add user form	☐	
PPL-02	Add user	Email, Name, Password, Role; submit	User appears in table	☐	
PPL-03	Role dropdown	Select each role	operator, technician, supervisor, tenant_admin saved correctly	☐	
PPL-04	Edit user	Click Edit; change name/role; Save	Changes persist	☐	
PPL-05	Delete user	Delete with confirm	User removed; cannot log in	☐	
PPL-06	Tenant admin access	Log in as tenant_admin	Can access <code>/people</code> for own org	☐	
PPL-07	Operator denied	Log in as operator	<code>/people</code> blocked or read-only per policy	☐	
PPL-08	Platform admin org selector	Log in as platform admin	Org selector switches tenant context	☐	

12. Platform admin — Tenants

ID	Test case	Steps	Expected result	Result	Notes
ADM-01	Admin page access	Platform admin → /admin/tenants	Tenants table + Create form	☐	
ADM-02	Non-admin blocked	Operator navigates to /admin/tenants	403 or redirect; page not accessible	☐	
ADM-03	Create tenant	Org code + Name; Create	Tenant in table	☐	
ADM-04	Enable CMMS	Click Enable CMMS on tenant	CMMS entitlement active; tenant sees embedded CMMS	☐	
ADM-05	Disable CMMS	Click Disable CMMS	CMMS shows disabled/external state for tenant	☐	
ADM-06	Link to People	Admin → People link for tenant	Opens /people?org={slug}	☐	

13. CMMS module

ID	Test case	Steps	Expected result	Result	Notes
CMMS-01	CMMS entitled — load	Tenant with CMMS enabled → /cmms	Embedded MooreviewCmms app loads	☐	
CMMS-02	CMMS disabled state	Tenant without entitlement	Disabled badge or external link message	☐	
CMMS-03	Overview tab stats	Open Overview	Open WOs, overdue PM, alarm WOs, PM WOs counts	☐	
CMMS-04	Work orders tab	Switch to Work orders	WO list and form visible	☐	
CMMS-05	Create work order	Fill title, asset, assignee, priority, due; Save	WO appears in list	☐	
CMMS-06	Edit work order	Change status / fields; Save	Updates persist	☐	
CMMS-07	Mark WO complete	Mark complete action	Status → complete; stats update	☐	
CMMS-08	Delete work order	Delete with confirm	WO removed	☐	
CMMS-09	PM schedules tab	Switch to PM schedules	PM list and form visible	☐	
CMMS-10	Create PM schedule	Title, asset, interval, next due; Save	PM appears in list	☐	
CMMS-11	Mark PM done	Mark PM done action	Next due date advances	☐	
CMMS-12	Generate due PM WOs	Click Generate due PM work orders	WOs created for due schedules	☐	

CMMS-13	Refresh lists	Refresh on WO/PM tabs	Data reloads from API	☐	
CMMS-14	Alarm → WO bridge	Trigger alarm with bridge enabled	Auto WO created (integration test)	☐	

14. Connectivity (/cellular/sims)

ID	Test case	Steps	Expected result	Result	Notes
CELL-01	Page loads	Tools → Connectivity or /cellular/sims	Alarm email, SMS, vendors, SIM sections	☐	
CELL-02	Alarm email config	Enter SMTP/SendGrid settings; Save	Config persists	☐	
CELL-03	Test alarm email	Click test send	Success or actionable error	☐	
CELL-04	Alarm SMS config	Enter Twilio settings; Save	Config persists	☐	
CELL-05	Test alarm SMS	Click test SMS	Success or actionable error	☐	
CELL-06	Add cellular vendor	Add Hologram / Twilio Super SIM	Vendor appears in list	☐	
CELL-07	Sync SIM inventory	Run sync	SIMs listed with status	☐	
CELL-08	Link SIM to Parc device	Link action	Association saved	☐	
CELL-09	Activate / deactivate SIM	Toggle SIM state	Status updates	☐	

15. Cloud sims (/cloud/sims)

ID	Test case	Steps	Expected result	Result	Notes
SIM-01	Page access	Navigate to /cloud/sims	Sim management UI or disabled state with instructions	☐	
SIM-02	Create sim	Name, Tenant ID, Type, Device ID, Interval	Sim instance created	☐	
SIM-03	Start sim	Click Start on instance	Sim runs; status updates	☐	
SIM-04	Stop sim	Click Stop	Sim stops	☐	
SIM-05	Delete sim	Delete with confirm	Instance removed	☐	

SIM-06	Feature disabled	MOOREVIEW_CLOUD_SIMS off	Disabled message with setup instructions	☐	
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16. Standalone pages

16a. I/O Map (/io-map)

ID	Test case	Steps	Expected result	Result	Notes
IOMAP-01	Page loads	Navigate to /io-map	I/O point panel + bindings sidebar	☐	
IOMAP-02	Enable I/O update	Toggle checkbox	Live I/O updates when enabled	☐	
IOMAP-03	Driver bindings	View bindings sidebar	Driver wiring and HMI screen bindings listed	☐	

16b. MV Draw (/mv-draw)

ID	Test case	Steps	Expected result	Result	Notes
MV-01	Page loads	Navigate to /mv-draw	2D grid canvas + toolbar	☐	
MV-02	New / Open / Save	File menu actions	Document lifecycle works	☐	
MV-03	Composer modes	Switch 2D / 3D / Plan	Mode changes; canvas updates	☐	
MV-04	Tools — Place / Connect	Place symbol; connect edges	Objects appear on canvas	☐	
MV-05	Undo / Redo	Perform edit; Undo; Redo	History stack works	☐	
MV-06	Compile HMI	Compile HMI from draw	HMI screens generated or errors shown	☐	
MV-07	Export package	Export package action	File downloads	☐	
MV-08	Unsaved changes alert	Navigate away with edits	Confirmation modal appears	☐	
MV-09	Responsive layout	Viewport ≤640px	Toolbar/layout adapts per breakpoints	☐	

17. Cloud Studio shared chrome

ID	Test case	Steps	Expected result	Result	Notes
CS-01	Brand header	Any Cloud Studio page	mooreVIEW brand visible	☐	
CS-02	Tenant slug label	Logged-in user on /sites	Correct tenant slug displayed	☐	
CS-03	Active nav highlight	Navigate between pages	Current page nav item highlighted	☐	
CS-04	Sign out — all pages	Sign out from People, Fleet, etc.	Consistent logout behavior	☐	
CS-05	Nav wrap — mobile	Viewport ≤720px	Nav wraps; all links reachable	☐	

18. Health, API & degraded states

ID	Test case	Steps	Expected result	Result	Notes
DEG-01	Health endpoint	GET /health	JSON status: deployment, runtime, dependencies	☐	
DEG-02	MongoDB unavailable	Simulate DB down (if possible)	UI shows graceful errors; no white screen	☐	
DEG-03	MQTT broker offline	Stop broker	Drivers/MQTT sections show offline; app usable	☐	
DEG-04	Empty data tables	Fresh tenant, no sites/devices	Empty states with guidance, not broken UI	☐	
DEG-05	API error handling	Force 500 on an API call	User-facing error message; no silent failure	☐	
DEG-06	Session expired	Invalidate session cookie; click action	Redirect to login with message	☐	

19. Cross-browser & accessibility smoke

ID	Test case	Steps	Expected result	Result	Notes
XBR-01	Chrome desktop	Run AUTH-02, SHELL-06, SITE-02	All pass on Chrome	☐	
XBR-02	Edge desktop	Run AUTH-02, HMI-01, FLEET-02	All pass on Edge	☐	

XBR-03	Firefox desktop	Run login + Studio load	No layout breakage	☐	
XBR-04	Keyboard navigation	Tab through login form	Focus order logical; Enter submits	☐	
XBR-05	Color contrast — login	Inspect login card text	Readable against Purple Standard background	☐	

20. End-to-end integration flows

ID	Test case	Steps	Expected result	Result	Notes
E2E-01	Platform onboarding	Admin creates tenant → adds user via People	New user can log in to Studio	☐	
E2E-02	Site pairing flow	Create site → copy code → configure edge agent	Agent online; cameras sync	☐	
E2E-03	Commissioning flow	System setup → Drivers → Tags → Program → HMI Apply	Project runs; HMI live	☐	
E2E-04	Operator day shift	Login → HMI → ack alarm → open camera → sign out	Full shift workflow without errors	☐	
E2E-05	CMMS maintenance	Enable CMMS → create PM → generate WO → complete	WO lifecycle complete	☐	
E2E-06	Fleet visibility	Add site + device + fleet asset	All three views consistent	☐	

Sign-off summary

Metric	Count
Total test cases	175
Passed	
Failed	
N/A	
Pass rate	

Open defects

ID	Severity	Summary	Ticket

Sign-off

Role	Name	Signature	Date
QA / Tester			
Product owner			
Engineering lead			